

Assignment 4

Due date: Apr 12th, 2026 before midnight

Problem 1. A manufacturing facility has two identical machines that operate independently. Each machine fails at a rate of 2 per month. There is a single repairperson who can repair one machine at a time, with repair rate 4 per month. When both machines are working, the system is in full production. When one machine is working and the other is being repaired, production continues at a reduced rate. When both machines are failed, production stops.

- Model the system as a Continuous-Time Markov Chain (CTMC). Clearly define the states and give the generator matrix Q .
- Find the steady-state probabilities $\tilde{\pi}_i$ for the CTMC. What is the expected production rate? (Assume production rate is 2 units/month when both machines work, 1 unit/month when one works, and 0 when none work.)
- Determine the embedded Markov Chain (EMC) for this system. Find the transition probability matrix P . Let π_i be the stationary distribution of the embedded DTMC.
- Verify that $\tilde{\pi}_i \propto \pi_i / \mu_i$, where $\mu_i = -q_{ii}$ is the total rate out of state i .

Solution

- Define states based on the number of failed machines:

- State 0: 0 failed machines (both working)
- State 1: 1 failed machine (one working, one being repaired or waiting)
- State 2: 2 failed machines (both failed, one being repaired, one waiting)

Transition rates:

- From state 0: failure rate 4 (to state 1)
- From state 1: failure rate 2 (to state 2), repair rate 4 (to state 0)
- From state 2: repair rate 4 (to state 1)

Generator matrix:

$$Q = \begin{pmatrix} -4 & 4 & 0 \\ 4 & -6 & 2 \\ 0 & 4 & -4 \end{pmatrix}$$

- Steady-state equations $\tilde{\pi}Q = 0$:

$$\begin{aligned} 4\tilde{\pi}_1 &= 4\tilde{\pi}_0 \\ 4\tilde{\pi}_0 + 4\tilde{\pi}_2 &= 6\tilde{\pi}_1 \\ \tilde{\pi}_0 + \tilde{\pi}_1 + \tilde{\pi}_2 &= 1 \end{aligned}$$

Solving: $\tilde{\pi}_0 = \frac{2}{5}$, $\tilde{\pi}_1 = \frac{2}{5}$, $\tilde{\pi}_2 = \frac{1}{5}$

Expected production rate:

$$E[\text{Production}] = 2\tilde{\pi}_0 + 1\tilde{\pi}_1 + 0\tilde{\pi}_2 = 2 \cdot \frac{2}{5} + 1 \cdot \frac{2}{5} = \frac{6}{5} = 1.2 \text{ units/month}$$

- (c) Embedded Markov Chain: $\mu_i = -q_{ii}$, $p_{ij} = q_{ij}/\mu_i$ for $i \neq j$:

$$\mu_0 = 4, \quad \mu_1 = 6, \quad \mu_2 = 4$$

$$P = \begin{pmatrix} 0 & 1 & 0 \\ \frac{2}{3} & 0 & \frac{1}{3} \\ 0 & 1 & 0 \end{pmatrix}$$

Stationary distribution $\pi = (\pi_0, \pi_1, \pi_2)$ satisfying $\pi P = \pi$:

$$\pi_0 = \frac{1}{3}, \quad \pi_1 = \frac{1}{2}, \quad \pi_2 = \frac{1}{6}$$

- (d) Verify $\tilde{\pi}_i \propto \pi_i/\mu_i$:

$$\frac{\pi_0}{\mu_0} = \frac{1/3}{4} = \frac{1}{12}, \quad \frac{\pi_1}{\mu_1} = \frac{1/2}{6} = \frac{1}{12}, \quad \frac{\pi_2}{\mu_2} = \frac{1/6}{4} = \frac{1}{24}$$

Normalizing:

$$\frac{1/12}{5/24} = \frac{2}{5} = \tilde{\pi}_0, \quad \frac{1/12}{5/24} = \frac{2}{5} = \tilde{\pi}_1, \quad \frac{1/24}{5/24} = \frac{1}{5} = \tilde{\pi}_2$$

$$\text{Thus } \tilde{\pi}_i = \frac{\pi_i/\mu_i}{\sum_j (\pi_j/\mu_j)}.$$

Problem 2. A small coffee shop has a single barista. Customers arrive according to a Poisson process at a rate of 10 per hour. The barista serves customers at an exponential rate of 12 per hour. However, customers are impatient: if a customer arrives and finds the barista busy, they will wait only for an exponentially distributed amount of time with mean 0.1 hour (6 minutes) before leaving the queue without being served. Once a customer starts being served, they never abandon.

- Model this system as a Continuous-Time Markov Chain (CTMC). Clearly define the states and give the generator matrix Q .
- Find the steady-state probabilities π_n for $n = 0, 1, 2, \dots$, where n is the number of customers in the system.
- What is the probability that an arriving customer abandons the queue?
- What is the expected number of customers in the system?
- What is the throughput (rate at which customers are served)?

Solution

- (a) Define states based on the number of customers in the system:

- State n : n customers in the system ($n = 0, 1, 2, \dots$)

Transition rates:

- Arrival rate: $\lambda = 10$ (from state n to $n + 1$)
- Service rate: $\mu = 12$ (from state n to $n - 1$ for $n \geq 1$)
- Abandonment rate: When there are n customers in the system, the number of waiting customers is $n - 1$ (for $n \geq 1$). Each waiting customer abandons at rate $\theta = 10$. Total abandonment rate from state n is $(n - 1)\theta$ for $n \geq 2$.

Generator matrix Q :

$$Q = \begin{pmatrix} -10 & 10 & 0 & 0 & 0 & \cdots \\ 12 & -(10+12) & 10 & 0 & 0 & \cdots \\ 0 & 12+10 & -(10+12+10) & 10 & 0 & \cdots \\ 0 & 0 & 12+20 & -(10+12+20) & 10 & \cdots \\ 0 & 0 & 0 & 12+30 & -(10+12+30) & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

(b) Steady-state equations $\pi Q = 0$:

For $n = 0$:

$$\lambda\pi_0 = \mu\pi_1 \Rightarrow 10\pi_0 = 12\pi_1 \Rightarrow \pi_1 = \frac{5}{6}\pi_0$$

For $n \geq 1$:

$$\lambda\pi_{n-1} + [\mu + n\theta]\pi_{n+1} = [\lambda + \mu + (n-1)\theta]\pi_n$$

This yields the recurrence:

$$\pi_{n+1} = \frac{\lambda}{\mu + n\theta}\pi_n = \frac{10}{12 + 10n}\pi_n \quad \text{for } n \geq 0$$

Thus:

$$\pi_n = \pi_0 \prod_{k=0}^{n-1} \frac{10}{12 + 10k} = \pi_0 \prod_{k=0}^{n-1} \frac{10}{10k + 12}$$

As $\sum_{n=0}^{\infty} \pi_n = 1$, we have

$$\pi_0 \left(1 + \frac{5}{6} + \frac{25}{66} + \frac{125}{1056} + \cdots \right) = 1$$

and thus $\pi_0 \approx 0.3679$.

(c) Throughput (successful service rate) $= \lambda(1 - P_{\text{abandon}}) = \mu(1 - \pi_0)$. Thus:

$$P_{\text{abandon}} = 1 - \frac{\mu(1 - \pi_0)}{\lambda} = 1 - \frac{12(1 - \pi_0)}{10}$$

With $\pi_0 \approx 0.3679$:

$$P_{\text{abandon}} = 1 - 1.2(0.6321) = 1 - 0.7585 = 0.2415$$

(d) Expected number of customers in the system:

$$L = \sum_{n=0}^{\infty} n\pi_n \approx 0 \cdot 0.3679 + 1 \cdot 0.3066 + 2 \cdot 0.1394 + 3 \cdot 0.0436 + 4 \cdot 0.0104 + \cdots \approx 0.96$$

(e) Throughput = rate of customers served $= \mu(1 - \pi_0) = 12(1 - 0.3679) = 12 \cdot 0.6321 = 7.585$ customers/hour.